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# **RV COLLEGE OF ENGINEERING**

**Bengaluru-560 059**

## **REPORT ON EXPERIENTIAL LEARNING**

**ACY 2025-26**

**CLUSTER**

**CSE**

**Title of the Project**

**Design and RTL Verification of a DDS-Based Embedded  
Signal Generator**

**Students Group**

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| <b>1.</b>      |                                                             |
| <b>2.</b>      |                                                             |

## 1. Introduction

Precise and stable waveform generation is a fundamental requirement in many embedded and digital systems such as communication systems, signal generators, motor controllers, audio devices, and electronic test instrumentation. In such applications, sine waves are particularly important because they form the basis of modulation, control algorithms, and signal analysis.

Traditionally, waveform generation in embedded systems is implemented using software running on microcontrollers or processors. However, software-based waveform generation often suffers from several limitations such as timing jitter, non-deterministic execution, limited frequency resolution, and dependence on processor load.

These limitations make software-only approaches unsuitable for high-precision and real-time signal generation tasks. To overcome these challenges, hardware-based signal generation techniques are preferred. One widely adopted digital technique is Direct Digital Synthesis (DDS), which enables accurate, stable, and repeatable waveform generation using purely digital hardware logic. DDS systems are commonly used in professional signal generators, radios, and communication equipment.

In this project, a DDS-based embedded signal generator is designed and verified using Verilog at the Register Transfer Level (RTL). The system generates a sinusoidal waveform digitally and converts it into a pulse-width modulated (PWM) signal that can be filtered to obtain an analog output. The complete design is verified through RTL simulation, which is the standard first step in embedded hardware development.

## 2. Problem Definition

### 2.1 Problem Statement

Embedded systems frequently require the generation of flexible and high-quality waveforms with stringent performance requirements. These requirements include accurate and fine-grained frequency control, stable and deterministic timing behavior, and minimal dependence on external analog components.

Conventional software-based waveform generation techniques suffer from inherent timing uncertainties caused by processor scheduling, interrupt handling, and varying computational workloads. Such non-deterministic behavior leads to timing jitter and limits the achievable precision of the generated waveforms. While the use of external

Digital-to-Analog Converters (DACs) can enhance signal accuracy, it also introduces additional system cost, increased power consumption, and greater hardware complexity.

Therefore, the fundamental problem addressed in this project is the design of a hardware-based waveform generation mechanism that is capable of producing a precise and stable sinusoidal signal using purely digital logic, while eliminating software-induced delays and avoiding reliance on external analog components.

## **2.2 Background Information (Literature Review)**

Direct Digital Synthesis (DDS) is a well-established and widely adopted technique for waveform generation in modern digital signal processing systems. DDS operates by digitally accumulating phase information and subsequently converting this phase into waveform amplitude values using lookup tables or algorithmic computation methods.

When compared to traditional analog oscillators and software-based waveform generation techniques, DDS offers several significant advantages. These include deterministic timing governed by a fixed system clock, exceptionally fine frequency resolution achieved through the use of high-precision phase accumulators, instantaneous frequency switching without transient effects, and highly repeatable and stable output characteristics.

DDS architectures are extensively documented in technical literature and application notes published by semiconductor manufacturers, and they are commonly implemented in Field Programmable Gate Array (FPGA) and System-on-Chip (SoC) based systems. In addition, Pulse Width Modulation (PWM) based digital-to-analog conversion is frequently employed in embedded systems as a cost-effective alternative to dedicated Digital-to-Analog Converters (DACs) when low-cost analog signal generation is required.

This project applies these established principles in an academic context by implementing a DDS-based signal generator using synthesizable Verilog at the Register Transfer Level (RTL) and validating its operation through standard open-source simulation and verification tools.

## **3. Objectives**

### **3.1 Primary Objectives**

- To design a DDS-based signal generator using pure digital hardware
- To implement the design using synthesizable Verilog RTL
- To generate a stable sinusoidal waveform with precise frequency control
- To eliminate software timing jitter by using clock-driven hardware logic

### **3.2 Secondary Objectives**

- To avoid the use of external DAC hardware
- To implement PWM-based digital-to-analog representation
- To verify the design through comprehensive RTL simulation
- To validate frequency accuracy using Python-based analysis
- To follow industry-standard embedded hardware design methodology

## **4. Methodology**

### **4.1 Approach:**

The methodology adopted in this project follows a systematic and structured embedded hardware design flow consistent with standard digital system design practices. The process begins with high-level architectural planning, where the functional requirements of the system—such as precise frequency control, deterministic timing behavior, and minimal dependence on external analog components—are identified. System constraints including clock frequency, resolution requirements, and design simplicity are analyzed and translated into a suitable hardware architecture based on the Direct Digital Synthesis (DDS) principle.

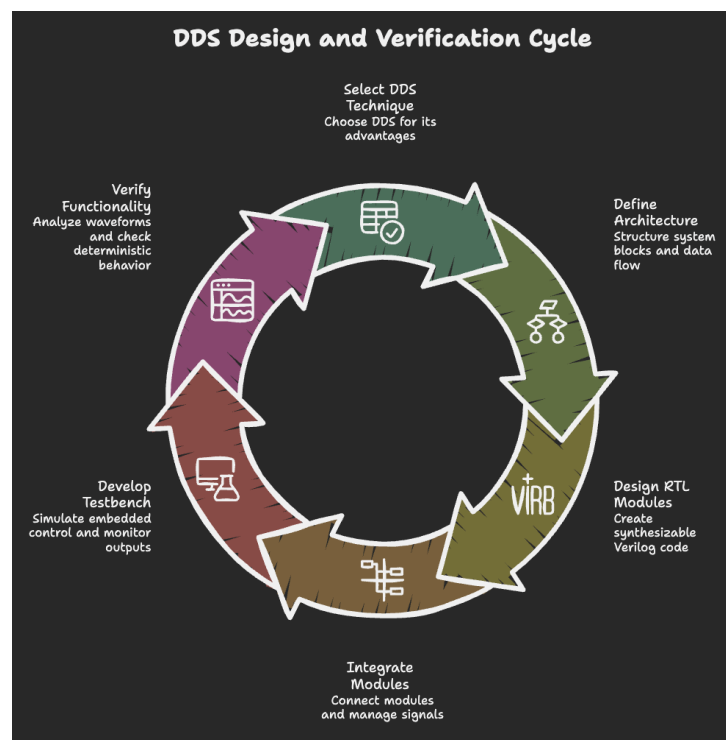
Following architectural definition, the system is decomposed into modular functional blocks, each responsible for a specific operation within the signal generation process. These blocks are designed and implemented using synthesizable Register Transfer Level (RTL) constructs in Verilog, ensuring that the design remains hardware-ready and portable across embedded platforms such as Field Programmable Gate Arrays (FPGAs) and System-on-Chips (SoCs). Emphasis is placed on modularity and clarity to facilitate independent development, easier debugging, and future scalability of the design.

Throughout the design process, deterministic operation and hardware efficiency are prioritized by employing clock-driven logic, fixed-point arithmetic, and lookup-table-based waveform generation. The methodology concludes with comprehensive RTL-level verification through simulation, which serves as the primary means of validating

functional correctness, timing behavior, and control logic. This simulation-first approach enables early detection of design issues and ensures that the final system is reliable and suitable for subsequent hardware implementation.

## 4.2 Procedures

The overall design and verification procedure is carried out through the following sequential steps:



- **Selection of Direct Digital Synthesis (DDS) as the Signal Generation Technique:**

DDS is chosen due to its deterministic timing behavior, fine frequency resolution, and suitability for fully digital hardware implementation.

- **Definition of System Architecture and Data Flow:**

The complete DDS system is decomposed into functional blocks, and the flow of data between the phase accumulator, sine lookup table, and PWM generator is clearly defined.

- **Design of Individual RTL Modules:**

Each functional block is independently designed using synthesizable Verilog RTL, ensuring clarity, modularity, and ease of verification.

- **Integration of Modules into a Complete System:**

The individual RTL modules are interconnected at the top level to form a complete DDS signal generator, with proper clocking, reset, and control signal management.

- **Development of a Testbench to Simulate Embedded Control:**

A comprehensive testbench is created to emulate the behavior of an embedded processor, allowing control parameters such as frequency tuning and enable signals to be applied and observed.

- **Verification of Functionality through Simulation and Waveform Analysis:**

The integrated system is verified using RTL simulation, and waveform analysis is performed to confirm correct phase accumulation, waveform generation, PWM modulation, and overall deterministic behavior.

## 5. Project Execution

### 5.1 Planning and Design

During the planning phase, the Direct Digital Synthesis (DDS) architecture was selected due to its proven capability to generate precise, stable, and deterministic waveforms using purely digital hardware. DDS was identified as an appropriate solution as it offers fine-grained frequency control, repeatable output behavior, and immunity to software-induced timing variations. To ensure clarity, maintainability, and ease of verification, the overall system was partitioned into modular hardware blocks, with each block performing a specific function within the signal generation pipeline.

The planned system architecture comprises:

- **Phase Accumulator** – responsible for digital phase generation based on the frequency tuning word.
- **Sine Lookup Table (LUT)** – converts phase information into corresponding waveform amplitude values.
- **Pulse Width Modulation (PWM) Generator** – provides digital-to-analog representation of the waveform by generating a duty-cycle-modulated output.
- **Control and Status Registers** – used for configuring system parameters and observing operational status.
- **Modular architecture** – enables independent development, testing, and verification of each component, thereby improving overall design robustness and scalability.

Key design parameters, including the bit width of the phase accumulator, the size of the sine LUT, and the resolution of the PWM generator, were carefully selected to achieve an optimal balance between frequency resolution, waveform accuracy, and hardware simplicity. These choices ensure that the design meets performance requirements while

remaining efficient and suitable for implementation on resource-constrained embedded hardware platforms.

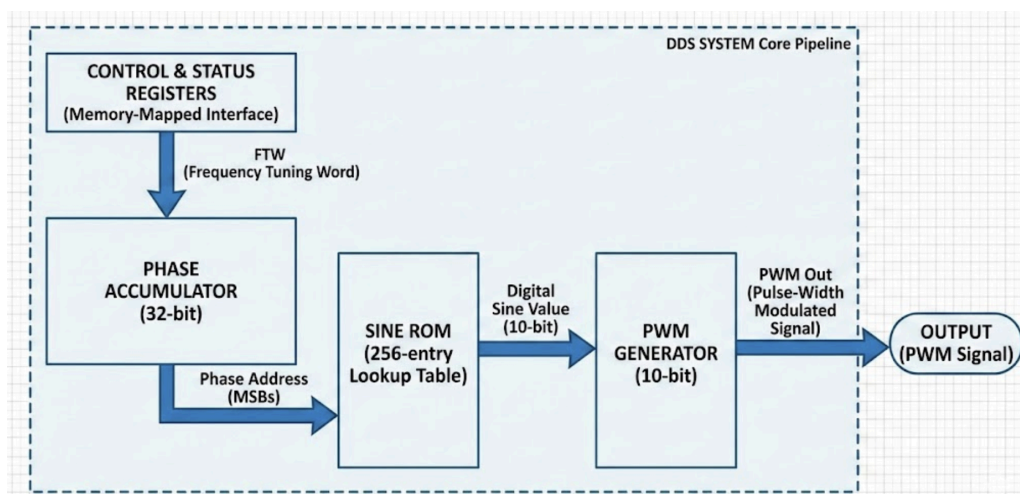
## 5.2 Implementation

The system was implemented using Verilog at the Register Transfer Level (RTL), with each functional block developed as an independent and self-contained module. These modules were subsequently integrated into a unified top-level design to form the complete DDS-based signal generation system. This modular implementation approach enhances design clarity, facilitates independent verification, and supports scalability and future system extensions.

Key implementation features of the design include:

- A 32-bit phase accumulator is implemented to achieve high frequency resolution and precise phase control.
- A 256-entry sine lookup table (LUT) with 10-bit amplitude resolution is used for accurate digital representation of the sinusoidal waveform.
- A 10-bit Pulse Width Modulation (PWM) generator is incorporated to provide digital-to-analog representation of the generated waveform.
- Memory-mapped control registers are included to enable programmable frequency tuning and system enable/disable control.

All modules were developed exclusively using synthesizable RTL constructs, ensuring that the design is fully hardware-realizable and suitable for synthesis and deployment on embedded hardware platforms such as Field Programmable Gate Arrays (FPGAs) and System-on-Chips (SoCs).



## 6. Tools and Techniques Used

## 6.1 Tools

- Icarus Verilog – Open-source Verilog simulator
- GTKWave – Waveform viewer for simulation analysis
- Python 3 – LUT generation and verification scripts
- Text editor / IDE – RTL and script development

## 6.2 Techniques

### • **Direct Digital Synthesis (DDS):**

Direct Digital Synthesis is employed as the core signal generation technique in this project. DDS enables precise and deterministic waveform generation by digitally accumulating phase information and converting it into waveform amplitude values. This approach provides fine frequency resolution, instantaneous frequency switching, and stable output characteristics, making it well suited for embedded and real-time signal generation applications.

### • **Register Transfer Level (RTL) Design:**

The entire system is designed at the Register Transfer Level, where the behavior of the hardware is described in terms of registers, combinational logic, and clocked data transfers. RTL design ensures that the system is synthesizable, hardware-realizable, and compatible with standard FPGA and SoC design flows.

### • **Fixed-Point Arithmetic:**

Fixed-point arithmetic is used throughout the design instead of floating-point computation to reduce hardware complexity and improve efficiency. By carefully selecting bit widths for phase, amplitude, and control values, the design achieves high precision while maintaining low resource utilization and deterministic timing behavior.

### • **Lookup-Table-Based Waveform Generation:**

A lookup-table-based approach is used to generate sine wave amplitudes from phase information. Pre-computed sine values stored in memory eliminate the need for real-time trigonometric computation, thereby reducing hardware complexity and enabling fast, deterministic waveform generation.

### • **PWM-Based Digital-to-Analog Conversion:**

Pulse Width Modulation is used to digitally represent analog waveform amplitudes without the need for an external Digital-to-Analog Converter. The duty cycle of the PWM signal is modulated according to the sine amplitude, allowing the average value of the digital signal to approximate an analog waveform when filtered.

- **Testbench-Driven Functional Verification:**

Functional verification is performed using a comprehensive Verilog testbench that simulates the behavior of an embedded controller. The testbench applies control inputs, observes outputs, and verifies correct system operation through simulation and waveform analysis, ensuring that the design meets functional and timing requirements prior to hardware implementation.

## 7. Partial Results

### 7.1 Initial Findings

During the initial phase of RTL simulation, the functional behavior of the core DDS blocks was carefully observed and analyzed using waveform inspection. The simulation results confirmed that the **phase accumulator** increments its value correctly on every clock cycle in accordance with the applied Frequency Tuning Word (FTW). This verified that the phase generation mechanism operates as intended and that the output frequency is accurately controlled by the programmed FTW value.

Further analysis showed that the **Lookup Table (LUT) addressing** correctly follows the phase progression generated by the accumulator. The most significant bits of the phase value cycle through the LUT address space in the expected sequence, resulting in a periodic and continuous traversal of sine amplitude values. This observation confirmed the correctness of the phase-to-amplitude mapping and validated the LUT access logic.

Additionally, the **Pulse Width Modulation (PWM) output** was observed to respond proportionally to changes in the sine amplitude values. Higher amplitude values produced wider PWM pulses, while lower amplitude values resulted in narrower pulses. This behavior demonstrated that the digital-to-analog representation using PWM was functioning correctly. Collectively, these initial findings validated the correctness of the core DDS functionality and established a stable baseline for further refinement.

### 7.2 Iterative Improvements

Based on insights gained from the initial simulation results, several refinements were introduced to improve waveform quality and overall system performance. First, the scaling of the sine LUT values was optimized to ensure full utilization of the available amplitude range. This adjustment allowed the waveform to span the complete resolution of the PWM output, improving signal dynamic range and accuracy.

Second, the **PWM resolution** was adjusted to achieve smoother waveform representation. By refining the PWM counter width and duty cycle mapping, quantization effects were reduced, resulting in a more accurate approximation of the analog sine waveform. This improvement was particularly evident when observing the averaged PWM output over time.

Finally, the **testbench coverage** was enhanced to strengthen verification confidence. Additional frequency test cases were introduced to validate correct DDS operation across a broader range of frequencies, including low and high-frequency scenarios. These iterative improvements ensured that the design not only functioned correctly under basic conditions but also remained robust and reliable across different operating configurations.

## 8. Results and Discussion

### 8.1 Final Results

The final DDS-based signal generation system achieved a high level of precision and reliability, as verified through comprehensive RTL simulation and waveform analysis. The use of a 32-bit phase accumulator resulted in a frequency resolution of approximately **0.0116 Hz**, enabling fine-grained control over the generated waveform frequency. Python-based verification confirmed **sub-parts-per-million (sub-PPM) frequency accuracy**, demonstrating the effectiveness of the DDS approach in achieving precise frequency synthesis.

The generated waveform exhibited **stable and fully deterministic behavior**, with all operations synchronized to the system clock and no observable timing jitter. Additionally, the Pulse Width Modulation (PWM) output accurately reflected the corresponding sine amplitude values, with duty cycle variations matching the expected waveform shape. These results collectively confirm the correct functional operation of the complete DDS signal generation pipeline.

### 8.2 Discussion

The observed results demonstrate that Direct Digital Synthesis is a highly effective technique for waveform generation in embedded systems requiring precision and deterministic timing. By implementing the entire signal generation process in hardware,

the design completely eliminates software-related timing uncertainties caused by interrupts, task scheduling, and processor workload variations.

Furthermore, the combination of a lookup-table-based sine generator and PWM-based digital-to-analog representation provides a **cost-effective alternative to external Digital-to-Analog Converters (DACs)**. This approach significantly reduces system complexity while maintaining acceptable waveform quality for many embedded applications. RTL-level simulation proved sufficient to validate the correctness of the architecture, functional behavior, and timing relationships, aligning with standard industry practices where simulation serves as the primary verification step prior to physical hardware implementation.

## **9. Prototype (Hardware/Software)**

### **9.1 Prototype Description**

The prototype developed in this project consists of a fully synthesizable Register Transfer Level (RTL) design that has been thoroughly verified through simulation. While no physical hardware implementation was carried out due to academic time constraints, the design strictly adheres to hardware-synthesizable coding practices. As a result, the system is hardware-ready and can be directly deployed on Field Programmable Gate Array (FPGA) platforms with minimal modification.

### **9.2 Development Process**

The development process followed a structured and iterative approach, which included the following key activities:

- Modular RTL coding, where each functional block was implemented as an independent and reusable module.
- Incremental integration of individual modules to progressively build the complete system and simplify debugging.
- Continuous simulation-based verification at each stage of development to ensure functional correctness and timing consistency.
- Python-based cross-validation of frequency accuracy to mathematically verify DDS behavior and confirm simulation results.

### 9.3 Testing and Validation

Testing and validation were performed using a comprehensive Verilog testbench designed to emulate embedded system control behavior. Multiple test cases were executed, including:

- Generation of sine waves at different frequencies to verify correct frequency tuning.
- Enable and disable control testing to confirm proper system activation and deactivation behavior.
- Dynamic frequency switching during runtime to ensure stable and deterministic operation under changing control conditions.

Waveforms captured using GTKWave were analyzed and served as visual evidence of correct functional operation and accurate signal behavior throughout the DDS system.

Terminal Output snippet:

[TC10] Testing UART Control...

[UART TB] Sending CMD: ADDR=0x00 DATA=0x000d1b71

[CSR] t=0: SWEEP STEP = 0

[CSR] t=0: FREQ\_WORD = 858993 (0x000d1b71)

[UART TB] Sending CMD: ADDR=0x18 DATA=0x00000001

[CSR] t=0: WAVE SEL = 1 (0=Sin,1=Tri,2=Sq,3=Saw)

## 10. Conclusion

### 10.1 Summary

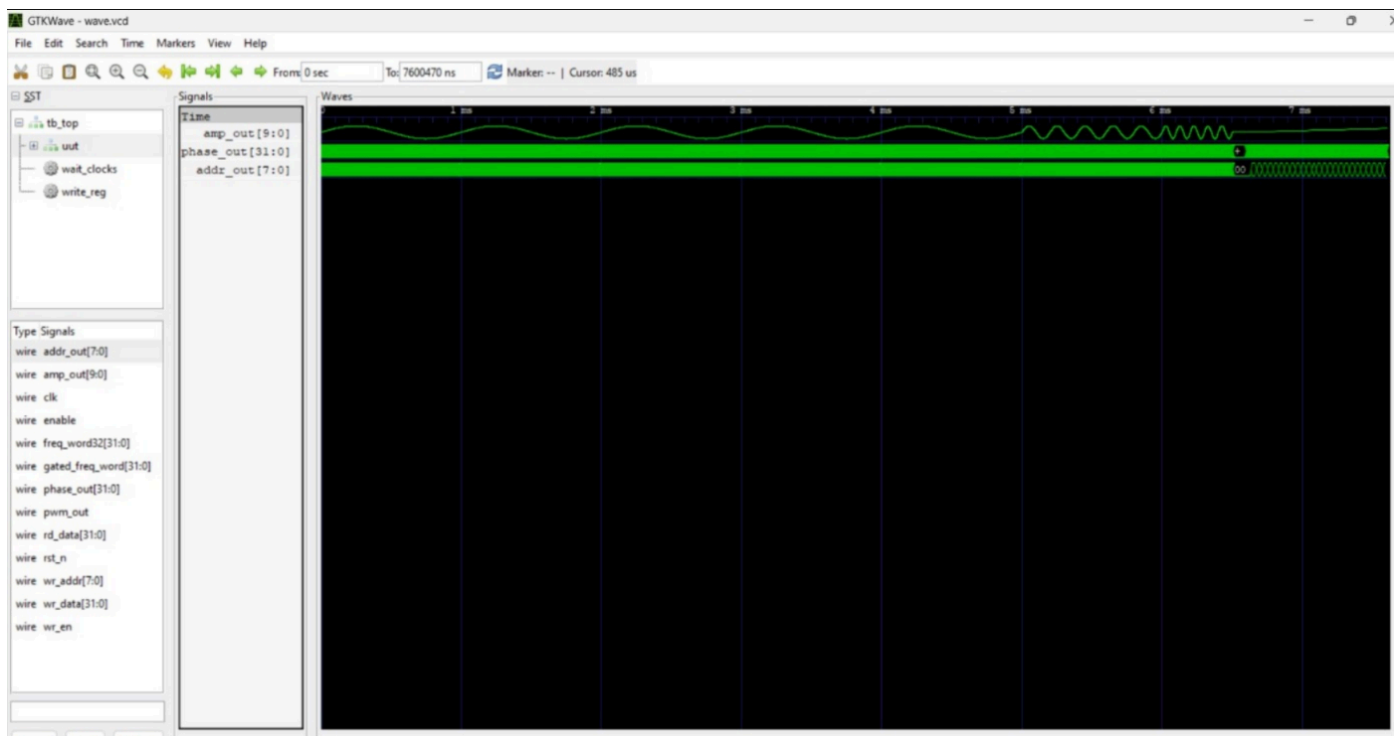
This project successfully demonstrates the design and RTL-level verification of a Direct Digital Synthesis (DDS) based embedded signal generator implemented using Verilog. By utilizing a phase accumulator for frequency control, a sine lookup table for waveform generation, and a PWM-based output stage for digital-to-analog representation, the system achieves precise, stable, and fully deterministic waveform generation using purely

digital hardware. Comprehensive RTL simulation and waveform analysis confirm the functional correctness and reliability of the proposed design.

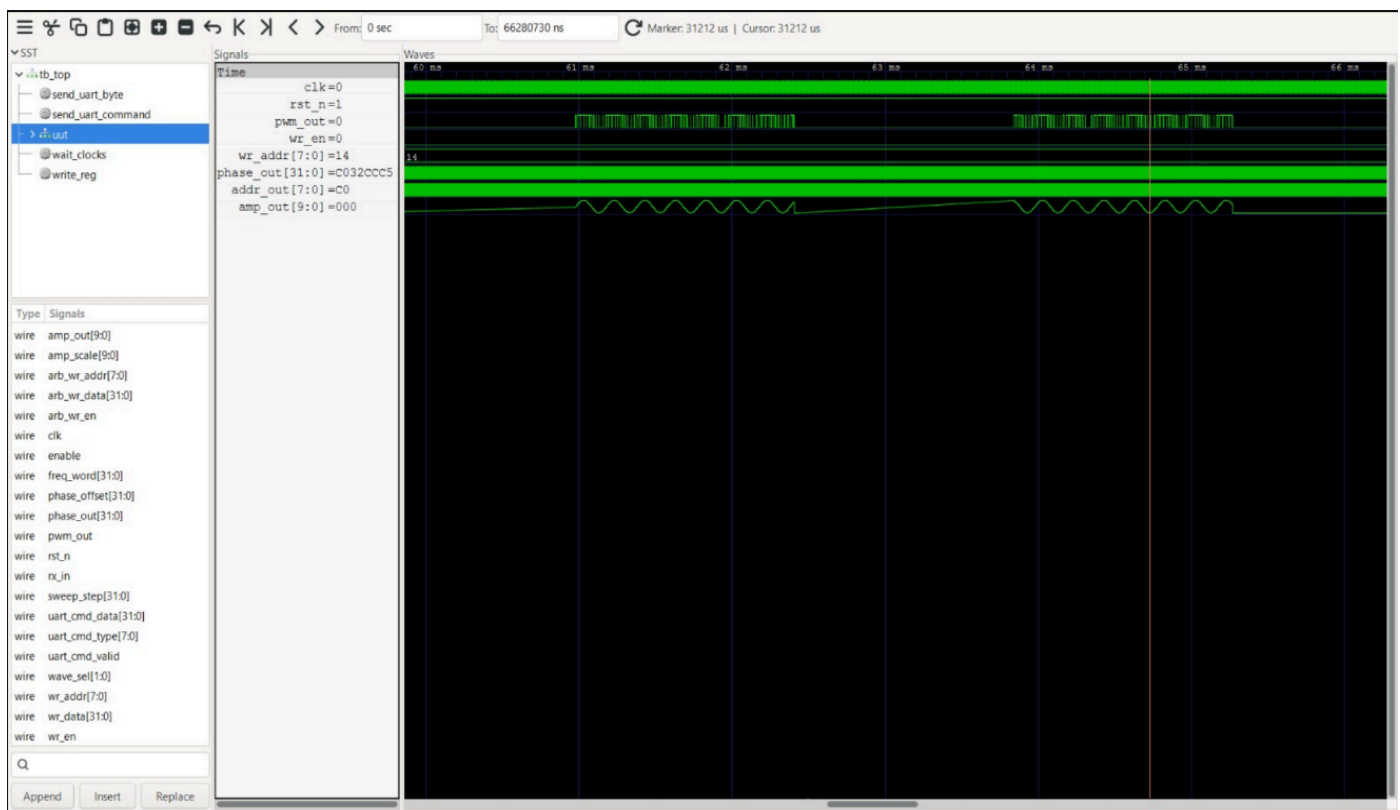
## 10.2 Personal Reflection

This project provided valuable hands-on experience in embedded hardware design, Register Transfer Level (RTL) coding, and simulation-based verification techniques. It enhanced practical understanding of digital signal generation concepts, fixed-point arithmetic, and modular hardware design principles, while also reinforcing familiarity with industry-standard embedded system development workflows and verification practices.

## 11. Visuals



**Fig.** Basic DDS functionality



**Fig.** DDS with advanced architecture integration

## 12. Outcome of the Work

The final outcome of the project includes:

- A fully synthesizable DDS-based signal generator
- Verified RTL design with comprehensive test coverage
- Sub-PPM frequency accuracy
- Hardware-ready architecture suitable for FPGA implementation
- Strong foundation for future extensions such as FPGA deployment, DAC integration, and advanced modulation feature

## 13. References

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